OreSat Reaction Wheel Upgrade

An Oregon Space Grant Consortium (OSGC) Undergraduate Team Experience Award Program (UTEAP) Project
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Introduction

Your satellite tumbles aimlessly tumbles unless you do something about it. Attitude Determination and Control System (ADCS) is doing something about it. Since space doesn't get you anything to push against or hold on to, you have to be creative. Most satellites control their orientation in space by a combination of reaction wheels, magnetorquers, and thrusters. OreSat uses the first two.

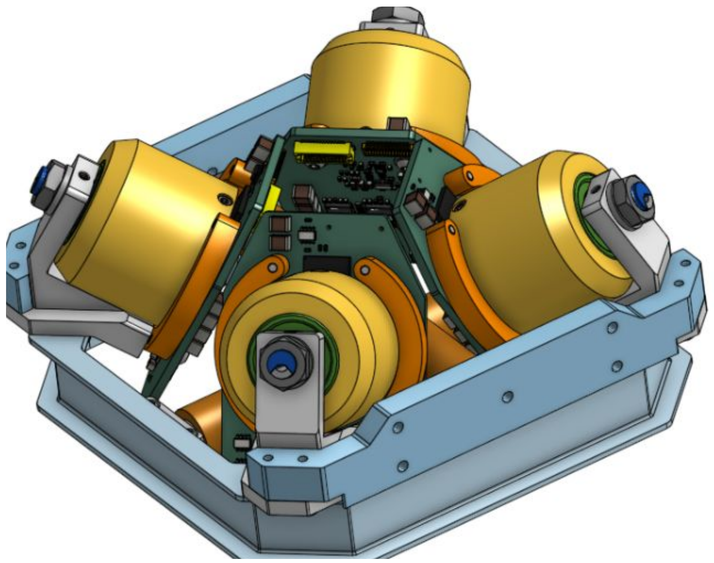


Figure: CAD of RW Assembly

By using 4 reaction wheel in the orientation shown here, we are able to manipulate the satellites in the x, y, and z axis independently to achieve the desired orientation of the satellite.

Reaction wheels are somewhat expensive. The OreSat team has been working to develop an open source approach to reaction wheels that is affordable. By using off the shelf components such as drone motors and hybrid bearings we've been able to get the cost of our reaction wheels down dramatically compared to commercial off the shelf options and with acceptable compromise to the quality of flight components.



Figure: Disassembled Drone Motor

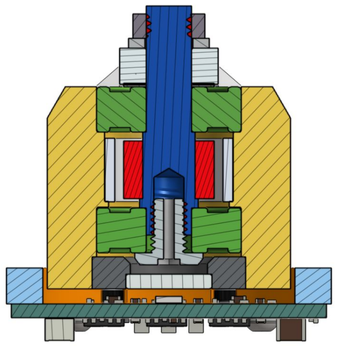


Figure: CAD Reaction Wheel

Previous Issues

We had no idea how to do this! Using hand tools to do hard things requires a certain finesse most students aren't capable of. By changing our standard operating procedures and revamping our design, we were able to successfully build and test flight capable units more often.



Figure: Reaction Wheels in assembly mock up



Figure: Completed reaction wheel

Success!!!

A completed reaction wheel ready to be integrated into the satellite's ADCS and hopefully make it to space. With UTEAP's generous grant we are able to fund our attempt at an affordable alternative to one of the most expensive and challenging problems that plague space flight. The work done by the OreSat team will go on to help teams around the world to achieve their own missions in space.

Acknowledgements

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This project could not have been completed without invaluable mentorship from Andrew Greenberg and Dr. James McNames.

Finally, thank you to the Oregon Space Grant Consortium for funding our improvements, and making accessibility a more important priority for CubeSat builders around the world.